

CHIMERA: A Multi-Level Representation Framework for Predicting Asymmetric Catalytic Reactions

Zongyong Hu^{a,d,e,1}, Jiebin Su^{a,1}, Jiaxing Liu^b, Chunjin Wei^c, Cheng Wang^{b,*}, Da Zhou^{a,**}

^a*School of Mathematical Sciences; National Institute for Data Science in Health and Medicine, Xiamen University, Xiamen, Fujian, 361005, China*

^b*State Key Laboratory for Physical Chemistry of Solid Surfaces, iChEM, College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, Fujian, 361005, China*

^c*School of Science, Jimei University, Xiamen, Fujian, 361021, China*

^d*Zhongguancun Academy, Beijing, 100080, China*

^e*Zhongguancun Institute of Artificial Intelligence, Beijing, 100080, China*

Abstract

Asymmetric catalytic reactions play a critical role in the synthesis of enantiomerically enriched compounds, with broad applications in pharmaceuticals and fine chemicals. While recent advances in machine learning have improved reaction prediction using molecular descriptors and graph-based representations, existing approaches still face challenges such as limited generalizability, poor interpretability, and the scarcity of high-quality, domain-specific datasets. In this work, we propose CHIMERA, a chirality-augmented and molecular interaction-enhanced multi-level representation framework designed to improve both predictive accuracy and interpretability in asymmetric catalysis. CHIMERA integrates molecular fingerprints, graph-based molecular structures, and motif-decomposed graph embeddings within a hierarchical architecture. It incorporates a chirality-aware attention mechanism and a molecular interaction module to highlight atom-level regions relevant to model prediction. We evaluate CHIMERA on two reaction datasets predominantly featuring oxazoline-based asymmetric catalysis. Experimental results show that CHIMERA achieves competitive predictive performance

*Corresponding author. E-mail: wangchengxmu@xmu.edu.cn

**Corresponding author. E-mail: zhouda@xmu.edu.cn

¹These authors contributed equally to this work.

and strong aggregate results across the evaluated datasets and tasks, offering a promising tool for data-driven catalyst design and optimization in asymmetric synthesis.

Keywords: Asymmetric catalysis; Molecular representation; Graph neural networks; Enantioselectivity prediction

1. Introduction

Asymmetric catalytic reactions play a vital role in modern chemical synthesis, particularly in pharmaceuticals, fine chemicals, and materials science. By employing chiral catalysts, these reactions enable the transformation of achiral or prochiral substrates into enantiomerically enriched products. Given that enantiomers often exhibit markedly different biological activity, reactivity, and physicochemical properties, the efficient and selective synthesis of enantiopure compounds has become a central goal in synthetic chemistry and drug development [1].

In asymmetric catalysis, traditional catalyst screening relies heavily on expert intuition and empirical trial-and-error, making the process time-consuming, costly, and inefficient [2]. In contrast, data-driven approaches leverage existing reaction datasets to train predictive models, significantly accelerating the identification of optimal catalysts and conditions, and improving reaction efficiency and selectivity [3, 4, 5].

Recent developments in machine learning have opened new avenues for modeling asymmetric catalysis. Early studies applied conventional machine learning methods—such as multivariate linear regression, XGBoost, and support vector machines (SVM)—to model catalytic processes [6, 7, 8, 9, 10, 11, 12, 3, 13, 14]. These methods typically rely on molecular descriptors as input and are trained on reaction-specific template datasets. For example, Sigman and colleagues constructed multidimensional regression models using structural and reaction parameters to predict enantioselectivity [12]. While effective, such models depend heavily on expert-defined features, limiting their ability to generalize across structurally diverse catalysts.

To overcome this limitation, recent efforts have turned to deep learning, which can automatically learn task-relevant features from raw molecular representations. In the context of asymmetric catalysis, deep learning offers the potential to identify enantioselective trends, uncover structure–selectivity relationships, and guide catalyst optimization without relying on handcrafted

features. Depending on input type, models can be broadly classified into sequence-based and graph-based approaches.

Sequence-based models use architectures such as recurrent neural networks (RNNs), long short-term memory (LSTM) networks, or Transformers to process molecular strings like SMILES. These models can capture complex molecular patterns and support rapid prediction for unseen compounds. In asymmetric catalysis, accurately capturing subtle chiral distinctions is essential for predictive modeling. Sequence-based models have demonstrated potential in effectively encoding chirality-related information embedded within SMILES representations, thereby facilitating improved prediction of enantioselective outcomes. For instance, Schwaller et al. treated reaction prediction as a language translation task using a sequence-to-sequence (Seq2Seq) model [14], while Gomez-Bombarelli et al. employed a variational autoencoder (VAE) to embed SMILES into a continuous latent space [15], demonstrating the ability of such models to learn meaningful molecular representations.

The rapid expansion of benchmark datasets and modeling strategies has also reshaped how enantioselectivity prediction is formulated and evaluated. Recent surveys and perspectives have consolidated developments from descriptor-based physical-organic models to deep molecular representations and data-efficient optimization, emphasizing that models for asymmetric catalysis should be assessed not only by predictive accuracy, but also by the choice of target transformation, applicability domain, local reaction-space coverage, and chemical interpretability [32, 33, 34, 35, 36]. In parallel, recent studies have demonstrated that reaction-level representations, catalyst-oriented fragment descriptors, and reaction-specific machine-learning models can predict enantioselective outcomes in organocatalytic and related asymmetric catalytic systems, even under limited-data conditions [37, 38, 13]. These advances highlight both the promise and the limitations of purely sequence- or descriptor-driven models.

However, linear SMILES encoding is insufficient to capture some stereochemical features (e.g. axial and planar chirality). In contrast, graph-based approaches offer a more chemically relevant representation of molecular structures. These models treat molecules as graphs and use graph neural networks (GNNs) to extract structural features. GNNs can directly encode topological and spatial relationships, leading to improved performance in complex property prediction tasks.

Moreover, GNN-based models are increasingly being tailored to reaction

performance prediction and asymmetric catalysis. DeepReac+ combines a graph-neural-network reaction encoder with active learning, directly using two-dimensional molecular structures to predict organic reaction outcomes and prioritize informative experiments [39]. Li et al. developed SEMG-MIGNN, in which steric- and electronic-information-embedded molecular graphs are coupled with a molecular interaction module to predict reaction yield and stereoselectivity with extrapolative ability and atom-level interpretability [16]. More recently, HCat-GNet was proposed as a human-interpretable GNN framework for ligand optimization in asymmetric catalysis, enabling enantioselectivity prediction from SMILES-defined reaction components and providing interpretable guidance for catalyst design [40]. Together, these studies show that graph-based learning has moved beyond generic reactivity prediction toward chemically interpretable modeling of asymmetric catalytic outcomes. By directly modeling molecular topology, local chemical environments, and interactions among reaction components, GNN-based methods bypass part of the dependence on handcrafted descriptors and offer powerful tools for asymmetric catalysis and related reaction-prediction tasks.

Despite these advances, several key challenges remain. First, many accurate models still depend on reaction-class-specific templates, quantum-chemical descriptors, or transition-state information, which can be computationally expensive and may restrict transferability across catalyst families. Second, most methods rely on a single molecular view—such as fingerprints, SMILES, descriptor vectors, or graphs—rather than integrating their complementary information. Finally, data scarcity remains a critical bottleneck, especially in asymmetric catalysis, where experimentally derived datasets are limited, heterogeneous, and often biased toward local reaction spaces; as a result, deep learning models may underperform when generalizing to new catalysts, substrates, or reaction conditions.

To address these challenges, we propose **CHIMERA—a Chirality-Augmented and Molecular Interaction-Enhanced Representation Algorithm** for asymmetric catalytic reactions. CHIMERA introduces a hierarchical framework that integrates molecular fingerprints, graph-based embeddings, and motif-decomposed substructures. It incorporates a molecular interaction module and a chiral-center-aware substructure extraction mechanism, and it provides model-derived atom-level saliency scores for post hoc interpretation. We curated two benchmark datasets of asymmetric catalytic reactions predominantly featuring oxazoline-based systems and conducted comprehensive

evaluations. Results demonstrate that CHIMERA achieves strong aggregate predictive performance within the evaluated reaction spaces, together with relative atom-level saliency visualizations.

2. MATERIALS AND METHODS

2.1. Dataset

This study focuses on enantioselectivity prediction for asymmetric catalytic reactions involving chiral ligand-metal catalyst systems, with many examples based on oxazoline-derived ligand scaffolds. Our objective is to predict the enantiomeric excess (ee) value of a reaction based on input variables, in order to optimize reaction conditions and catalyst design, thereby improving the enantioselectivity of catalytic processes. Enantiomeric excess value is a widely used metric to quantify the ratio of two enantiomers in a chiral compound. It is commonly used to evaluate the efficiency of asymmetric synthesis or chiral separation, and to assess the selectivity of a chiral catalyst. The ee value is defined as:

$$\text{ee (\%)} = \frac{|R - S|}{R + S} \times 100\% \quad (1)$$

where R and S represent the concentrations or yields of the two enantiomers. In practical modeling scenarios, to improve the stability and accuracy of predictions, the enantiomeric excess (ee) value is typically transformed into the differential Gibbs free energy ($\Delta\Delta G$) for regression [12]. The conversion is defined as:

$$\Delta\Delta G = -RT \ln \left(\frac{1 + \text{ee}}{1 - \text{ee}} \right) \quad (2)$$

where R is the gas constant, T is the temperature in Kelvin, and ee is expressed as a decimal. This transformation allows for a more physically meaningful and numerically stable modeling of enantioselectivity.

We curated two asymmetric catalysis datasets with different degrees of chemical heterogeneity. Data A contains 3,646 cleaned reaction records covering diverse, predominantly C-C bond-forming asymmetric transformations with N,N-chelating, N,N,N-tridentate, and related chiral ligand classes, 21 metal centers mainly represented by Ni, Cu, and Pd, and broad variation in substrates, products, solvents, temperatures, and reaction times.

Data B contains 886 photoredox/Ni-catalyzed asymmetric reaction records, including diverse chiral ligands, substrates, products, Ni sources, solvents, additives/bases, and photocatalysts. Compared with Data A, Data B represents a more focused reaction-family dataset while still retaining substantial structural and condition diversity.

The reaction records in both datasets were curated from publicly available reaction records and original literature reports. When available, the released data files retain the corresponding bibliographic or source information to facilitate traceability. Detailed summaries of Data A and Data B are provided in Tables 1 and 2, respectively. To further characterize the target distributions, we plotted the ee-value histograms of the two datasets, as shown in Figure 1. The variables actually used as model inputs differ between the two datasets and are specified in Section 2.3.3.

Table 1: Summary of Data A (Multi-System Dataset)

Variable	Description
Ligand Information	Diverse chiral ligand classes, including oxazoline-derived, N,N-chelating, N,N,N-tridentate, and related ligand scaffolds
Metal Center	Types of metal centers used in catalytic reactions, mainly including Ni, Cu, Pd, and other metals
Reactants and Products	Reactants and their corresponding products across diverse asymmetric transformations
Solvent	Types of solvents employed in the reactions
Reaction Conditions	Parameters such as temperature and reaction time
Enantiomeric Excess Value	Quantitative indicators of reaction efficiency and stereoselectivity
Total number of data entries: 3646	

2.2. Model Architecture

CHIMERA is a multi-level representation algorithm tailored for asymmetric catalysis, augmented by chiral features and molecular interactions. Figure 2 illustrates its overall workflow. The abbreviations used in Figure 2 are defined as follows. CHASE denotes the Chiral-center-Aware Substructure Extractor, a self-attention-based module that extends the MSAN-SE

Table 2: Summary of Data B (Focused Reaction-Family Dataset)

Variable	Description
Ligand Information	Predominantly oxazoline-derived or bis-oxazoline chiral ligands, with related ligand variants
Metal Center	Ni sources used in photoredox/Ni-catalyzed asymmetric reactions
Reactants and Products	Reactants and their corresponding products within a focused reaction family
Solvent	Types of solvents employed in the reactions
Reaction Conditions	Parameters such as temperature and reaction time
Additional Additives	Photocatalysts, additives/bases, and other auxiliary components used in the reactions
Enantiomeric Excess Value	Quantitative indicators of reaction efficiency and stereoselectivity
Total number of data entries: 886	

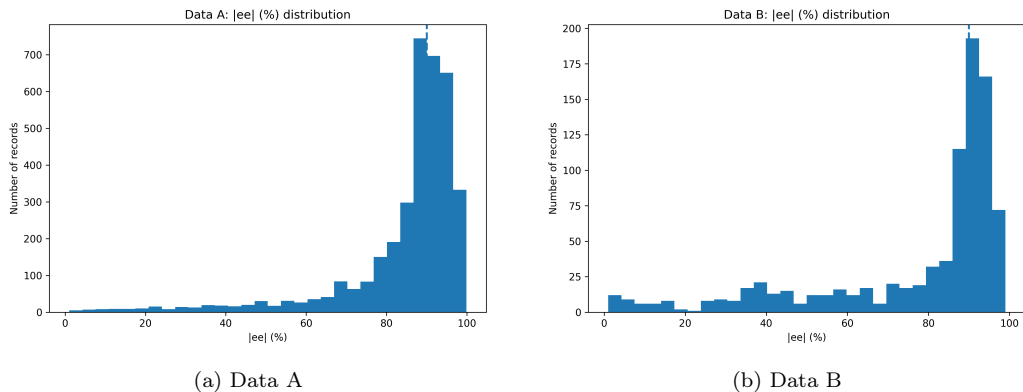


Figure 1: Distribution of ee values in the two datasets. Left: Data A. Right: Data B.

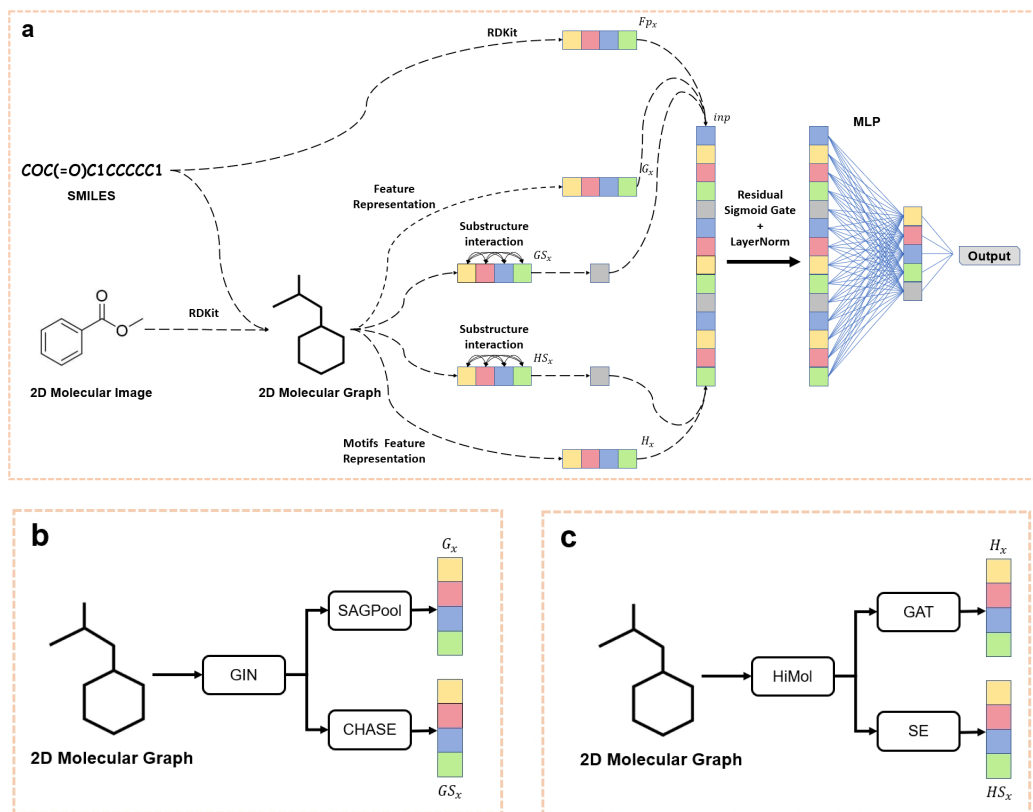


Figure 2: Workflow of CHIMERA. a. RDKit is used to extract SMILES-based and 2D molecular graph fingerprint representations (Fp_x). From the 2D graph, structural features (G_x) and motif-based features (GS_x , H_x , and HS_x) are further derived. Molecular interaction features are computed from GS_x and HS_x and concatenated into a unified feature vector (inp). The fused vector is adaptively reweighted by a residual sigmoid gating module, normalized with LayerNorm, and then passed to a task-specific multilayer perceptron prediction head. b. Workflow of feature representation. c. Workflow of motif feature representation.

For Data A, only molecular-structure-based representations are used as model inputs. For Data B, the molecular representation is additionally concatenated with an auxiliary molecular-descriptor vector and encoded reaction-condition variables, which are omitted from the schematic for clarity.

substructure extraction mechanism by assigning additional weights to chiral centers during substructure-pattern learning [31]. HiMol denotes Hierarchical Molecular Graph Self-supervised Learning; in CHIMERA, we adopt its hierarchical molecular graph construction to encode atom-, motif-, and graph-level representations [23]. SE denotes Substructure Extraction, referring to the non-chirality-weighted substructure extraction branch used to obtain motif-level substructure representations. SAGPool denotes Self-Attention Graph Pooling, which scores and pools graph nodes according to node features and graph topology [20]. For each reaction component—including reactants, catalysts, and products—we first obtain SMILES strings, molecular graph representations, and motif-decomposed molecular graphs. RDKit is an open-source cheminformatics library that is widely used for molecular modeling, feature extraction, and molecular manipulation. Local topological and stereochemical features are extracted from SMILES using RDKit’s fingerprint functions. A Graph Isomorphism Network (GIN) captures latent local and global structural features from molecular graphs, while a Graph Attention Network (GAT) processes motif-decomposed graphs to extract functional substructures embedded within molecules. Additionally, a molecular interaction module identifies key interactions at the substructure level between molecules. Finally, CHIMERA integrates these diverse, cross-modal feature representations to predict outcomes of asymmetric catalytic reactions.

2.2.1. Molecular Fingerprint

Molecular fingerprints typically encode local structural information, such as atomic neighborhoods and functional groups, by mapping them into fixed-length binary or integer vectors. By inputting the SMILES representations of the product and ligand into the RDKit fingerprint generation function, their extended connectivity fingerprint (ECFP) vectors can be obtained. These vectors are then concatenated to form a comprehensive reaction feature vector [17].

$$v_i = f_I(t_i), \quad i \in \{\text{product}, \text{ligand}\} \quad (3)$$

$$v_{\text{fp}} = [v_{\text{product}}, v_{\text{ligand}}] \quad (4)$$

where t_i is the corresponding SMILES representation, f_I is the RDKit fingerprint generation function, and v_{fp} captures global structural information from the product and ligand.

Table 3: Details of initial molecular embeddings

Features	Descriptions	Dimensions
Atomic Features		
Atom symbol	One-hot encoding of atomic symbols	35
Atom degree	One-hot encoding of the number of non-hydrogen neighbors	6
Formal charge	One-hot encoding of formal charge	8
Hybridization	One-hot encoding of orbital hybridization types	7
In ring	Binary feature indicating whether the atom belongs to a ring	1
Aromaticity	Binary feature indicating whether the atom is in an aromatic system	1
Chirality type	One-hot encoding of specific chirality types	4
Bond Features		
Bond type	One-hot encoding of bond types (single, double, triple, aromatic)	4
Conjugation	Binary feature indicating whether the bond is conjugated	1
In ring	Binary feature indicating whether the bond is in a ring structure	1
Stereochemistry	One-hot encoding of bond stereochemistry	4

2.2.2. Molecular Graph-Based Feature Representation

Molecular graphs more accurately capture the relationships between atoms and chemical bonds. In this section, we utilize cheminformatics tools such as RDKit to convert the SMILES strings of each molecule involved in the reaction into their corresponding 2D molecular graphs, denoted as $G = (V, E)$. Each node V is represented by a node feature vector X_v , which consists of a set of atomic features. Each edge E is represented by an edge feature vector E_{km} composed of bond features, where the subscript km indicates that atoms k and m are bonded. The complete initial numerical features of the molecules are summarized in Table 3.

We employ the GIN model along with jumping knowledge (JK) pooling to obtain comprehensive node-level representations [18, 19]. Additionally, we introduce the SAGPool module, based on a self-attention mechanism, to enhance and pool node features [20].

$$h_v^{(k)} = MLP^{(k)} \left((1 + \epsilon^{(k)}) \cdot h_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)} \right) \quad (5)$$

$$H_{JK} = concat_{JK} (\{H^{(k)}\}) \quad (6)$$

$$H_G = SAGPool(H_{JK}, A) \quad (7)$$

Here, $h_v^{(k)}$ denotes the feature representation of node v at the k -th layer, $\mathcal{N}(v)$ represents the set of neighbors of v , $\epsilon^{(k)}$ is a learnable parameter or predefined constant at the k -th layer, and $MLP^{(k)}$ refers to the multilayer perceptron at the same layer. The symbol $concat_{JK}$ denotes the Jumping Knowledge concatenation-aggregation operation, and $SAGPool$ refers to the Self-Attention Graph Pooling module based on the self-attention mechanism. H_G represents the global representation of the entire molecular graph. SAGPool additionally provides node scoring values that reflect the relative emphasis assigned to nodes in the pooled graph representation.

2.2.3. Motif-Based Decomposed Feature Representation

To capture the local functional substructures embedded within molecular structures [21, 22], we adopt a motif-based molecular representation method based on Hierarchical Molecular Graph Self-supervised Learning (Hi-Mol) [23]. The decomposition of the original molecular graph G is extended from the BRICS algorithm by further fragmenting macrocyclic components

and selecting the smallest rings as the resulting motifs [24].

To further incorporate global information, we introduce a graph-level node V_g into the molecular graph. This node connects to all motif nodes through a set of graph-level edges E_g , enabling the aggregation of global molecular information via direct connections.

On the constructed augmented graph $\tilde{G}$, we employ a Graph Attention Network (GAT) to encode and obtain representations at the atom, motif, and graph levels [25]. Specifically, at the l -th layer, the representation of each node is updated using the following aggregation formula:

$$h_v^{l+1} = \sum_{j \in \mathcal{N}(v) \cup \{v\}} \alpha_{v,j} \Theta_l h_j^l \quad (8)$$

Here, h_v^l denotes the representation of node v at the l -th layer, with the initial representation $h_v^{(0)} = X_v^a$, i.e., the atomic features of node v . $\mathcal{N}(v)$ represents the set of neighboring nodes of v ; Θ_l is the learnable weight matrix at layer l ; and $\alpha_{v,j}$ denotes the attention coefficient between node v and its neighbor j .

Through multi-layer GAT encoding, the features of all node types in the augmented graph are iteratively updated, yielding atom-level representations $H_a \in \mathbb{R}^{|V| \times d}$, motif-level representations $H_m \in \mathbb{R}^{k \times d}$, and graph-level representations $H_g \in \mathbb{R}^d$. These hierarchical representations collectively form a comprehensive molecular graph embedding, which can be employed for subsequent enantioselectivity prediction and other downstream tasks.

2.2.4. Molecular Interaction Module

In asymmetric catalytic reactions, the interactions between different molecular species critically influence enantioselectivity [26, 27, 28, 29, 30]. To model and capture these interaction patterns among key substructures across molecules, we introduce a Molecular Interaction Module. Building upon the self-attention mechanism of MSAN-SE, we apply additional weighting to chiral centers, enabling the model to emphasize stereochemically relevant nodes during substructure pattern extraction [31].

We define a chiral-center indicator matrix $C \in \mathbb{R}^{M \times N}$, where M denotes the number of learnable substructure-pattern query vectors, N denotes the number of molecular nodes, and d denotes the embedding dimension. Each element of C indicates the presence or importance of a chiral center at the

corresponding node position:

$$C_{m,i} = \begin{cases} \alpha, & \text{if node } i \text{ is a chiral center} \\ 1, & \text{otherwise} \end{cases}$$

Here, $\alpha > 1$ is an amplification factor. In the CHASE branch, this indicator matrix is integrated into the attention score computation:

$$A_{pre} = \frac{QK^T}{\sqrt{d}}, \quad (9)$$

$$A = \text{softmax}(A_{pre} \odot C), \quad (10)$$

$$O = Q + AV. \quad (11)$$

where softmax denotes the normalization function, A_{pre} is the preliminary attention score matrix, and $\odot$ represents the Hadamard (element-wise) product. Q denotes the projected learnable query-pattern matrix, while K and V denote the projected key and value representations of molecular nodes. The resulting substructure representation is denoted as $O = [x_1, x_2, \dots, x_M] \in \mathbb{R}^{M \times d}$, where M is the number of learnable pattern vectors and d is the embedding dimension.

For each molecular pair, interaction matrices are constructed from two complementary substructure-representation branches: the chirality-aware CHASE branch and the non-chirality-weighted SE branch. For each branch $b \in \{\text{CHASE}, \text{SE}\}$, we compute the interaction matrices between the ligand representation l and the product or reactants $j \in \{pro, r_1, r_2\}$:

$$S_{mn}^{(b;l,j)} = \frac{\left(x_{m,b}^{(l)}\right)^T x_{n,b}^{(j)}}{\left\|x_{m,b}^{(l)}\right\|_2 \left\|x_{n,b}^{(j)}\right\|_2}, \quad m, n = 1, \dots, M. \quad (12)$$

The six interaction matrices are subsequently flattened and concatenated:

$$v_{\text{int}} = [\text{flatten}(S^{(\text{CHASE};l,pro)}), \text{flatten}(S^{(\text{CHASE};l,r_1)}), \text{flatten}(S^{(\text{CHASE};l,r_2)}), \\ \text{flatten}(S^{(\text{SE};l,pro)}), \text{flatten}(S^{(\text{SE};l,r_1)}), \text{flatten}(S^{(\text{SE};l,r_2)})] \in \mathbb{R}^{6M^2}. \quad (13)$$

2.2.5. Enantioselectivity Prediction of Reactions

To effectively integrate multi-level representations of reaction data, we concatenate the feature vectors from different sources into a fused reaction representation, denoted as $inp \in \mathbb{R}^{d'}$. A residual sigmoid gate is then applied to adaptively reweight the fused features:

$$g(inp) = \frac{1}{2} + \sigma(\text{MLP}(inp)), \quad (14)$$

$$z = \text{LayerNorm}(g(inp) \odot inp), \quad (15)$$

where σ denotes the sigmoid function and $\odot$ denotes element-wise multiplication. The additive offset shifts the range of the multiplicative gate to $(0.5, 1.5)$, allowing individual feature dimensions to be adaptively attenuated or enhanced. Layer normalization is subsequently applied to stabilize the reweighted representation, and the resulting feature vector z is used for downstream prediction.

For regression, a task-specific multilayer perceptron head $G_{\text{reg}}(\cdot)$ is used to predict the differential Gibbs free energy ($\Delta\Delta G$):

$$\hat{y} = G_{\text{reg}}(z), \quad (16)$$

where $\hat{y}$ denotes the predicted $\Delta\Delta G$ value, as defined in Equation 2. When needed, the predicted $\Delta\Delta G$ can be converted back to the corresponding ee value according to Equation 2. The regression model is optimized using the Mean Squared Error (MSE) loss:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2, \quad (17)$$

where N denotes the number of training samples, and $\hat{y}_i$ and y_i are the predicted and ground-truth values of the i -th sample, respectively.

For classification, another task-specific multilayer perceptron head $G_{\text{cls}}(\cdot)$ is used to predict the class probability distribution:

$$\hat{p} = G_{\text{cls}}(z) \in \mathbb{R}^C, \quad (18)$$

where $\hat{p}$ denotes the predicted probability distribution over C classes. The classification model is optimized using the Cross-Entropy (CE) loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log \hat{p}_{i,c}, \quad (19)$$

where $y_{i,c}$ is the true label of the i -th sample for class c (typically one-hot encoded), and $\hat{p}_{i,c}$ is the predicted probability for class c .

2.3. Evaluation Metrics

2.3.1. Regression Tasks

In regression tasks, the model performs numerical prediction of $\Delta\Delta G$ to further assess its ability to accurately fit the reaction outcomes. To measure the error between predicted and true values, we employ the following two evaluation metrics:

1. Root Mean Square Error (RMSE)

RMSE reflects the average deviation between predicted and actual values. It is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (20)$$

where y_i is the ground truth value of the i -th sample, $\hat{y}_i$ is the corresponding predicted value, and n denotes the total number of samples. RMSE is sensitive to large errors, making it effective at capturing the model’s performance on extreme values.

2. Coefficient of Determination (R^2)

R^2 measures the proportion of variance in the dependent variable that is predictable from the independent variables. It is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (21)$$

where $\bar{y}$ is the mean of the true values. A higher R^2 value (closer to 1) indicates that the model better explains the variance in the data.

2.3.2. Classification Tasks

In classification tasks, the model aims to predict discrete variables representing reaction outcomes. Specifically, enantiomeric excess values are categorized into two classes based on a threshold of 90: reactions with values ≥ 90 are assigned to the high enantiomeric excess class, while those with lower values are assigned to the low enantiomeric excess class. This threshold was selected based on the empirical ee distributions shown in Figure 1, where 90% ee provides a practical separation between high- and lower-enantioselectivity reactions while avoiding a severely imbalanced classification setting. Chemically, reactions with ee values around or above 90% are commonly regarded

as highly enantioselective. To evaluate model performance in classification, we adopt the following commonly used metrics:

1. Recall (REC)

Recall measures the proportion of actual positive samples correctly identified by the model. It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (22)$$

where TP is the number of true positives and FN is the number of false negatives. Recall reflects the model’s ability to detect positive instances.

2. F1 Score (F1)

The F1 Score is the harmonic mean of precision and recall, providing a balanced measure in scenarios of class imbalance:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (23)$$

where Precision is defined as the proportion of true positives among all positive predictions. A higher F1 Score indicates a more balanced trade-off between precision and recall.

3. Area Under the Curve (AUC)

AUC denotes the area under the Receiver Operating Characteristic (ROC) curve, ranging from 0 to 1. The ROC curve illustrates the relationship between the false positive rate (FPR) and the true positive rate (TPR) across different classification thresholds. Compared to other metrics, AUC is more robust to class imbalance. It provides a comprehensive assessment of overall model performance by evaluating its discriminative ability across all possible thresholds. A higher AUC value, closer to 1, indicates stronger capability of the model to distinguish between positive and negative samples.

2.3.3. Experimental Settings

The four final CHIMERA configurations use the same core optimization settings. Models are trained with the Adam optimizer for 200 epochs using a batch size of 128, an initial learning rate of 0.001, and L_2 weight decay of 1×10^{-4} . A `ReduceLROnPlateau` scheduler reduces the learning rate by a factor of 0.1 after 10 epochs without improvement in the *validation loss*, with a minimum learning rate of 1×10^{-7} . All 200 epochs are executed; early stopping is not used. The final architecture uses a node embedding dimension of 32, five GIN layers, two GAT layers, 10 learnable interaction patterns, a

residual sigmoid fusion gate with hidden dimension 128, LayerNorm after feature reweighting, and a prediction head with hidden dimensions 1024 and 256 and dropout 0.1. The chiral-center amplification factor is fixed at $\alpha = 2.0$ for all four dataset–task combinations. All experiments were conducted using a single NVIDIA A100 GPU.

For evaluation, each dataset is partitioned into 10 outer folds using seed 42. For classification, stratified 10-fold splitting is used when class counts permit; regression uses shuffled 10-fold splitting. When outer fold k is the test fold, the next outer fold in cyclic order is used in its entirety as the validation fold, and the remaining eight folds form the training set, giving a fixed 8/1/1 train/validation/test fold allocation. The validation fold is used exclusively for learning-rate scheduling and checkpoint selection: classification retains the epoch with the highest validation AUC, whereas regression retains the epoch with the lowest validation RMSE. The outer-test samples are excluded from training and validation, and their outcome labels are not accessed during any training epoch, scheduler update, or checkpoint decision. After all 200 epochs are complete, the validation-selected checkpoint is loaded and the outer test fold is evaluated once.

The same outer-fold assignments and leakage-free evaluation protocol were used for all baseline and ablation experiments. For classical baseline models that do not require validation-based checkpoint selection, the validation fold was nevertheless excluded from model fitting to ensure the same training-data allocation across methods. The same validation-only model-selection principle was also applied in the component-based splitting experiments.

The final input composition differs between Data A and Data B because the availability and consistency of reaction-condition annotations vary across the two datasets. For Data A, CHIMERA uses molecular-structure-based inputs only. Specifically, the fingerprint branch is constructed from the product and ligand SMILES, where the ligand is obtained from the `Ni_ligan_Smiles` field, whereas the graph and motif branches process the ligand, product, R1, and R2 molecular graphs. Reaction-condition fields available in the Data A files, including temperature, reaction time, and metal identity, are retained as metadata for compatibility with the shared data pipeline but are not included in the final model representation. Accordingly, the fused input for Data A is defined as

$$inp^{(A)} = v_{\text{mol}}, \quad (24)$$

where v_{mol} denotes the concatenated molecular representation before feature reweighting.

For Data B, the molecular representation is additionally concatenated with an auxiliary molecular-descriptor vector and an encoded reaction-condition vector:

$$\text{inp}^{(B)} = [v_{\text{mol}} \parallel v_{\text{desc}} \parallel c_{\text{cond}}], \quad (25)$$

where $\parallel$ denotes vector concatenation, v_{desc} denotes the additional atom- and motif-level molecular descriptor vector, and c_{cond} contains temperature, reaction time, metal source, solvent, additive/base, photocatalyst, and irradiation-condition information.

3. Results

3.1. Performance Comparisons

To evaluate predictive performance and robustness, we employ a unified 10-fold cross-validation protocol on both datasets. Each dataset is partitioned into ten approximately equal outer folds. In each iteration, one fold is held out as the test set, the next fold in cyclic order is used as the validation set, and the remaining eight folds are used for model training. The validation fold is used exclusively for learning-rate scheduling and checkpoint selection, whereas the held-out test fold is evaluated only after model selection is complete. The final results are reported as the mean and standard deviation over the ten held-out test folds, thereby reducing dependence on any single data partition and providing a more stable estimate of predictive performance within the curated reaction space.

In addition to random record-level cross-validation, we evaluate CHIMERA under component-based partitioning strategies in which reactions are split according to the catalyst, product, reactant 1, or reactant 2. These experiments provide a more stringent assessment of generalization when specific reaction components are absent from the corresponding training folds. As summarized in Table 4, component-based splitting leads to a clear deterioration relative to random record-level splitting, indicating that extrapolation to held-out reaction components remains substantially more challenging than interpolation within the reaction space represented during training. Among the four splitting strategies, product-based splitting consistently yields the strongest performance on both datasets. For Data A, the product split achieves an

AUC of 0.794 and an R^2 of 0.301, whereas the corresponding values for Data B are 0.778 and 0.338, respectively. In contrast, catalyst-based splitting is the most challenging setting, yielding AUC values of 0.584 and 0.689 and mean R^2 values of -0.040 and -0.308 on Data A and Data B, respectively. Reactant-based splits generally exhibit intermediate performance. The relatively large fold-to-fold variation observed for several component-based splits, particularly on Data B, further indicates that predictive difficulty depends strongly on the specific reaction components assigned to the held-out folds. Overall, these results identify generalization to unseen catalyst environments as a major remaining challenge for CHIMERA and, more broadly, for data-driven enantioselectivity prediction.

Table 5 summarizes the model comparisons for two prediction tasks: binary high-ee/low-ee classification, evaluated using recall (REC), F1 score, and area under the receiver operating characteristic curve (AUC), and continuous $\Delta\Delta G$ regression, evaluated using the coefficient of determination (R^2) and root mean squared error (RMSE). The classification task predicts the magnitude category of enantioselectivity and does not predict its sense, namely whether the major product has the *R* or *S* configuration.

The baseline methods cover three principal categories. The first category consists of classical descriptor- and fingerprint-based machine-learning methods, including DESC_SVM, which combines hand-crafted molecular descriptors with a support vector machine, and ECFP_GB, which uses extended-connectivity fingerprints together with gradient boosting. The second category includes standard graph neural networks, namely GCN, GIN, and GAT, which encode molecular structures through graph convolution, graph-isomorphism-based aggregation, and attention-based message passing, respectively. The third category comprises reaction- or chirality-aware deep-learning approaches. RXNFP [41] is a Transformer-based model pre-trained on large-scale reaction corpora to generate reaction-level representations; CHIENN [42] incorporates stereochemical information through edge-based ordering of local neighborhoods; and MFP_MLP concatenates molecular fingerprints and processes the resulting representation using a multilayer perceptron.

The comparison results in Table 5 show that CHIMERA achieves competitive performance across the two datasets and prediction tasks. On the more heterogeneous Data A, CHIMERA obtains the numerically highest mean values for the five reported metrics, with a REC of 0.787, an F1 score of 0.788, an AUC of 0.864, an R^2 of 0.445, and an RMSE of 0.728. However, the mar-

gins over the strongest competing methods are generally modest, particularly for classification. For regression, CHIMERA shows a somewhat larger numerical difference: compared with DESC_SVM, R^2 increases from 0.423 to 0.445 and RMSE decreases from 0.741 to 0.728; compared with MFP_MLP, R^2 increases from 0.404 to 0.445 and RMSE decreases from 0.755 to 0.728. These results indicate that the multi-level representation used by CHIMERA can provide additional predictive information beyond global fingerprints or hand-crafted descriptors in this dataset, although the magnitude of the improvement varies across metrics.

On Data B, the differences among the leading models are smaller. CHIMERA achieves a REC of 0.773, ties RXNFP for the highest F1 score at 0.770, and obtains an AUC of 0.857, which is essentially comparable to the 0.858 achieved by RXNFP. For regression, CHIMERA yields an R^2 of 0.650 and an RMSE of 1.572, compared with 0.646 and 1.585 for MFP_MLP, respectively. Thus, while CHIMERA remains competitive across both prediction tasks, the numerical differences relative to the strongest baselines are limited on this more focused reaction-family dataset.

To further assess the uncertainty in these differences, we performed paired fold-level comparisons between CHIMERA and MFP_MLP using the identical ten outer-test folds. On Data A, the mean paired AUC difference was 0.004 (95% CI: -0.013 to 0.020), while the mean paired differences for regression were 0.041 in R^2 (95% CI: 0.012 to 0.069) and 0.027 in RMSE reduction (95% CI: 0.008 to 0.047). On Data B, the corresponding differences were 0.009 for AUC (95% CI: -0.014 to 0.032), 0.003 for R^2 (95% CI: -0.037 to 0.043), and 0.013 for RMSE reduction (95% CI: -0.079 to 0.105). After Holm correction, none of the six two-sided Wilcoxon signed-rank comparisons reached the conventional significance threshold of 0.05. These results therefore do not support a claim of uniform statistically significant superiority over MFP_MLP, and instead indicate that CHIMERA provides competitive performance with generally modest differences relative to a strong fingerprint-based baseline.

It is also noteworthy that the standard graph-based models do not consistently outperform fingerprint- or descriptor-based approaches. On Data A, for example, GCN, GIN, and GAT achieve regression R^2 values of 0.269, 0.312, and 0.234, respectively, compared with 0.445 for CHIMERA. A similar pattern is observed on Data B. These results suggest that molecular topology alone may not capture all information relevant to the subtle structure-selectivity relationships in asymmetric catalysis. CHIMERA combines global

fingerprint information with atom-level graph representations, motif-level structural patterns, and intermolecular interaction features, providing a unified multi-level representation that remains competitive across the evaluated datasets and prediction tasks.

Table 4: Performance Comparison of CHIMERA with Different Splitting Strategies

Datasets	Split Methods	Classification						Regression			
		REC		F1		AUC		R^2		RMSE	
		mean	std	mean	std	mean	std	mean	std	mean	std
Data A	By Catalyst	0.528	0.045	0.423	0.129	0.584	0.115	-0.040	0.175	0.972	0.167
	By Product	0.710	0.039	0.707	0.042	0.794	0.047	0.301	0.083	0.813	0.028
	By Reactant 1	0.693	0.058	0.688	0.065	0.761	0.067	0.244	0.114	0.838	0.038
	By Reactant 2	0.696	0.044	0.694	0.048	0.760	0.050	0.229	0.103	0.850	0.100
Data B	By Catalyst	0.604	0.133	0.502	0.177	0.689	0.158	-0.308	0.871	2.537	0.596
	By Product	0.701	0.067	0.673	0.081	0.778	0.090	0.338	0.336	1.982	0.660
	By Reactant 1	0.605	0.094	0.565	0.116	0.674	0.095	0.037	0.336	2.215	0.720
	By Reactant 2	0.675	0.091	0.624	0.105	0.705	0.116	-0.056	0.768	2.178	0.737

3.2. Ablation Analysis

To investigate the contribution of the major representation components in CHIMERA, we conducted structural ablation experiments by removing one component at a time while retaining the remaining architecture and training protocol. Four variants were considered: (1) w/o Interaction, in which the molecular interaction representation was removed; (2) w/o Motifs, in which the motif-based decomposed representation was excluded; (3) w/o Fingerprint, in which the molecular fingerprint branch was omitted; and (4) w/o Atom-level Graph, in which the atom-level molecular graph representation was removed. The results are summarized in Table 7.

On Data A, the complete CHIMERA model achieves the strongest regression performance, with an R^2 of 0.445 and an RMSE of 0.728. Removing the interaction, motif, or atom-level graph representation leads to modest deterioration in regression performance, whereas removing the fingerprint branch results in a substantially larger decrease, reducing R^2 from 0.445 to 0.309 and increasing RMSE from 0.728 to 0.813. For classification, however, the contribution of individual learned structural representations is less monotonic. The w/o Interaction, w/o Motifs, and w/o Atom-level Graph variants achieve classification results comparable to, and in several metrics slightly

Table 5: Performance Comparison of CHIMERA with Baseline Methods

Datasets	Models	Classification						Regression			
		REC		F1		AUC		R^2		RMSE	
		mean	std	mean	std	mean	std	mean	std	mean	std
Data A	DESC_SVM	0.751	0.015	0.752	0.015	0.822	0.019	0.423	0.086	0.741	0.055
	ECFP_GB	0.763	0.024	0.765	0.024	0.853	0.018	0.343	0.025	0.793	0.035
	GCN	0.738	0.024	0.738	0.025	0.820	0.028	0.269	0.075	0.836	0.057
	GIN	0.764	0.024	0.764	0.024	0.842	0.018	0.312	0.041	0.811	0.038
	GAT	0.741	0.020	0.741	0.020	0.826	0.023	0.234	0.063	0.855	0.046
	CHIENN	0.720	0.029	0.719	0.032	0.815	0.025	0.205	0.080	0.873	0.072
	RXNFP	0.775	0.013	0.776	0.013	0.856	0.018	0.374	0.058	0.774	0.049
	MFP_MLP	0.783	0.021	0.783	0.022	0.860	0.016	0.404	0.043	0.755	0.040
	CHIMERA (Ours)	0.787	0.026	0.788	0.027	0.864	0.020	0.445	0.069	0.728	0.045
Data B	DESC_SVM	0.758	0.036	0.749	0.040	0.840	0.029	0.532	0.094	1.823	0.218
	ECFP_GB	0.758	0.045	0.755	0.046	0.843	0.029	0.632	0.092	1.613	0.197
	GCN	0.737	0.053	0.733	0.055	0.842	0.034	0.529	0.126	1.820	0.276
	GIN	0.742	0.039	0.737	0.043	0.836	0.040	0.543	0.139	1.788	0.297
	GAT	0.726	0.026	0.723	0.027	0.840	0.032	0.515	0.132	1.843	0.258
	CHIENN	0.739	0.034	0.731	0.037	0.824	0.031	0.530	0.109	1.821	0.216
	RXNFP	0.770	0.021	0.770	0.021	0.858	0.035	0.578	0.126	1.719	0.239
	MFP_MLP	0.764	0.042	0.760	0.044	0.848	0.034	0.646	0.056	1.585	0.119
	CHIMERA (Ours)	0.773	0.037	0.770	0.039	0.857	0.033	0.650	0.092	1.572	0.219

Table 6: Paired Fold-Level Statistical Comparison between CHIMERA and MFP_MLP

Dataset	Task	Metric	Mean paired difference (unadjusted 95% CI)	Holm-adjusted Wilcoxon p
Data A	Classification	AUC	0.004 [−0.013, 0.020]	1.000
Data A	Regression	R^2	0.041 [0.012, 0.069]	0.117
Data A	Regression	RMSE	0.027 [0.008, 0.047]	0.117
Data B	Classification	AUC	0.009 [−0.014, 0.032]	1.000
Data B	Regression	R^2	0.003 [−0.037, 0.043]	1.000
Data B	Regression	RMSE	0.013 [−0.079, 0.105]	1.000

Note: Positive differences favor CHIMERA. Differences are defined as CHIMERA minus MFP_MLP for AUC and R^2 , and as MFP_MLP minus CHIMERA for RMSE. Confidence intervals are unadjusted 95% Student- t intervals for the mean paired difference. Reported p -values are from two-sided Wilcoxon signed-rank tests with Holm correction across the six comparisons.

higher than, those of the complete model. In contrast, removing the fingerprint representation produces a clear decline across all three classification metrics, with AUC decreasing from 0.864 to 0.818.

A similar but more task-dependent pattern is observed on Data B. The complete model achieves the highest recall and F1 score among the evaluated configurations, reaching 0.773 and 0.770, respectively. Removing the interaction or motif representation reduces both metrics, while removal of the atom-level graph representation results in only a small decrease in recall and F1 and a slight increase in AUC from 0.857 to 0.864. For regression, the w/o Interaction variant performs slightly worse than the complete model, whereas the w/o Motifs variant achieves marginally better performance ($R^2 = 0.654$ and RMSE= 1.559 compared with $R^2 = 0.650$ and RMSE= 1.572 for the complete model). Removing the atom-level graph representation produces nearly unchanged regression performance. In contrast, removing molecular fingerprints again causes the clearest degradation, reducing R^2 to 0.550 and increasing RMSE to 1.783.

Overall, the ablation results indicate that the contributions of the different representation branches are complementary but task dependent rather than uniformly additive across all evaluation metrics. Molecular fingerprints provide the most consistent contribution across both datasets, particularly for continuous $\Delta\Delta G$ prediction. The interaction, motif-level, and atom-level graph representations provide additional structural information whose benefits vary with the dataset and prediction objective, while partial redundancy among these learned representations is evident in several settings. Taken together, these results support the use of the complete CHIMERA architecture as a unified multi-level representation framework that maintains competitive performance across different datasets and prediction tasks.

3.2.1. Sensitivity of the Chirality Amplification Factor

To examine robustness to the chiral-center amplification factor α independently of the fixed setting used in the main comparisons, we evaluated $\alpha \in \{1.5, 2.0, 2.5, 3.0\}$. The same scalar is assigned to every detected chiral-center node, whereas non-chiral nodes retain a weight of 1. As shown in Figure 3, the performance curves exhibit only modest variation over the tested range, indicating that CHIMERA is not highly sensitive to a single manually chosen value of α . Moreover, no single value of α consistently optimizes every metric across both datasets and both tasks.

Table 7: Results of CHIMERA Ablation Analysis

Datasets	Models	Classification						Regression			
		REC		F1		AUC		R^2		RMSE	
		mean	std	mean	std	mean	std	mean	std	mean	std
Data A	CHIMERA (Ours)	0.787	0.026	0.788	0.027	0.864	0.020	0.445	0.069	0.728	0.045
	w/o Interaction	0.794	0.020	0.795	0.021	0.869	0.017	0.424	0.085	0.741	0.059
	w/o Motifs	0.791	0.022	0.791	0.022	0.872	0.014	0.437	0.088	0.732	0.058
	w/o Fingerprint	0.738	0.024	0.738	0.024	0.818	0.024	0.309	0.059	0.813	0.054
	w/o Atom-level Graph	0.790	0.018	0.791	0.018	0.868	0.015	0.423	0.078	0.742	0.054
Data B	CHIMERA (Ours)	0.773	0.037	0.770	0.039	0.857	0.033	0.650	0.092	1.572	0.219
	w/o Interaction	0.761	0.039	0.753	0.045	0.859	0.034	0.643	0.086	1.587	0.191
	w/o Motifs	0.756	0.035	0.753	0.037	0.851	0.035	0.654	0.102	1.559	0.215
	w/o Fingerprint	0.757	0.024	0.753	0.025	0.855	0.035	0.550	0.121	1.783	0.285
	w/o Atom-level Graph	0.768	0.042	0.764	0.043	0.864	0.031	0.650	0.082	1.574	0.194

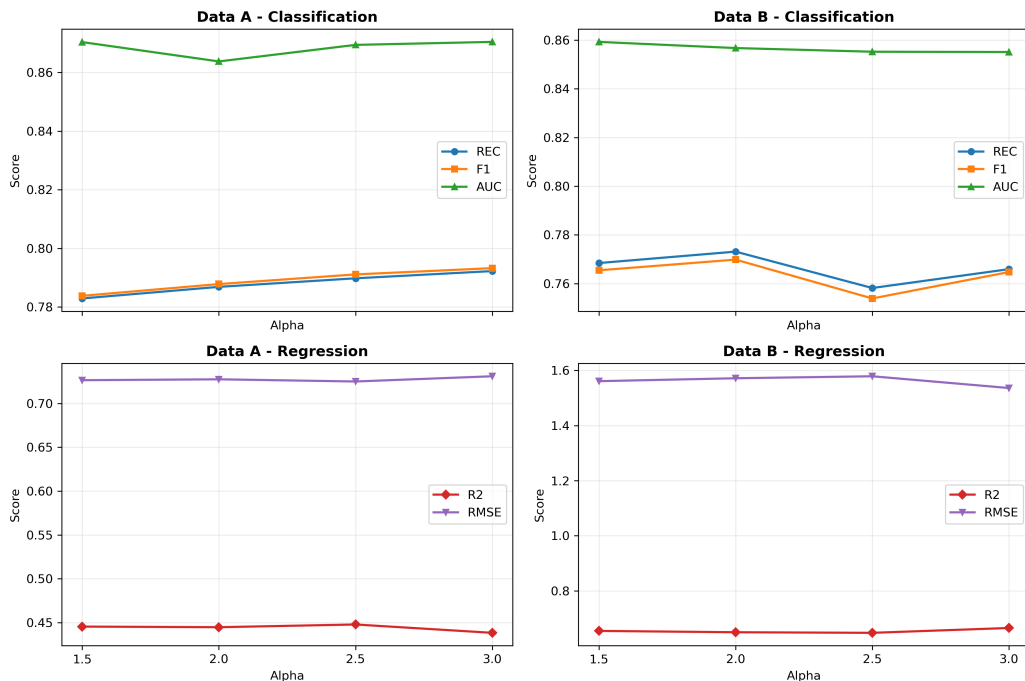


Figure 3: Sensitivity analysis of the chiral-center amplification factor α on Data A and Data B for classification and regression tasks.

3.3. Model-Derived Atom-Level Saliency Analysis

The SAGPool module provides model-derived atom-level saliency scores that can be used for post hoc visualization. These scores enable visualization of molecular regions receiving different levels of model attention during prediction. By leveraging the attention mechanism within the SAGPool module, CHIMERA generates model-derived atom-level attention scores as an intermediate visualization signal, enabling intuitive visualization of molecular regions that receive different levels of attention during prediction. The attention-based node scoring and pooling mechanism of SAGPool is defined by the following equations:

$$y = \text{GNN}(X, A) \quad (26)$$

$$i = \text{topk}(y) \quad (27)$$

$$X' = (X \odot \tanh(y))_i \quad (28)$$

$$A' = A_{i,i} \quad (29)$$

Here, X and A denote the feature matrix and adjacency matrix of the molecular graph, respectively. y is the attention score vector for the nodes obtained from the GNN. The top- k nodes are selected via $\text{topk}(y)$, and their features are modulated by $\tanh(y)$ using element-wise multiplication. A' is the sub-graph adjacency matrix corresponding to the selected node indices i .

To provide an illustrative external-case analysis, we conducted validation on three reaction predictions using external datasets, as shown in Figure 4. Additionally, we visualized the molecular structures by leveraging the attention maps, as illustrated in Figure 5. Specifically, for each molecule, the top 20% of atoms ranked by attention scores were highlighted in red, providing an intuitive representation of the atomic regions receiving relatively high attention during model prediction.

In Figure 5, it can be observed that most alkyl groups attached to the benzene ring are highlighted in red, indicating that these regions receive relatively high attention from the model. One possible chemical interpretation is that these regions may be associated with structural or electronic features favorable to the reaction. When the metal center approaches the ligand, the hyperconjugation system formed between the alkyl groups and the benzene ring may facilitate coordination with the metal. Such an electronic effect could strengthen the interaction between the metal and the ligand and thereby influence reaction reactivity and selectivity.

This pattern is particularly evident in Reaction 1, where the fluorine ion and bromine atom receive relatively high attention scores. One possible interpretation is that these atoms may be associated with local electronic environments relevant to the reaction. In addition, oxygen atoms commonly serve as coordination sites in ligands, forming stable coordination bonds with transition metals through their lone electron pairs. Such coordination may stabilize the catalyst–metal complex structure and influence the geometry of the transition state, thereby affecting the reaction pathway and enantioselectivity. This interpretation is also consistent with Reaction 1, where the N and O atoms in the ligand receive relatively high attention scores and are located near a chemically plausible coordination region.

On the other hand, the chiral centers in chiral catalysts can induce distinct three-dimensional configurations, which may affect the binding mode between the catalyst and the metal center. Due to the spatial distortion associated with chiral centers, symmetrically equivalent atoms within the same catalyst may experience different steric and electronic environments, resulting in different local interactions with the metal center. This pattern can also be observed in Figure 5, where symmetrically equivalent atoms exhibit different attention intensities. One possible interpretation is that the model captures chirality-related differences in the local electronic and steric environments of these atomic regions.

Taken together, the SAGPool-based attention maps provide a model-derived visualization of atom-level saliency. In the present datasets, several highlighted regions are qualitatively consistent with chemically plausible stereoelectronic and coordination features of the studied ligand–metal catalytic systems. This correspondence suggests that the saliency maps may reflect chemically relevant structural patterns within the evaluated reaction space and can provide a reference for chemically informed interpretation and hypothesis generation. Further studies involving broader catalyst families and reaction classes are needed to determine whether these attention patterns generalize beyond the reaction spaces considered here.

4. Conclusion and Discussion

In this study, we introduce CHIMERA, a multi-level representation algorithm designed for asymmetric catalysis. By integrating multi-level reaction representations, CHIMERA incorporates molecular interaction modeling and chirality-aware attention mechanisms to improve predictive performance

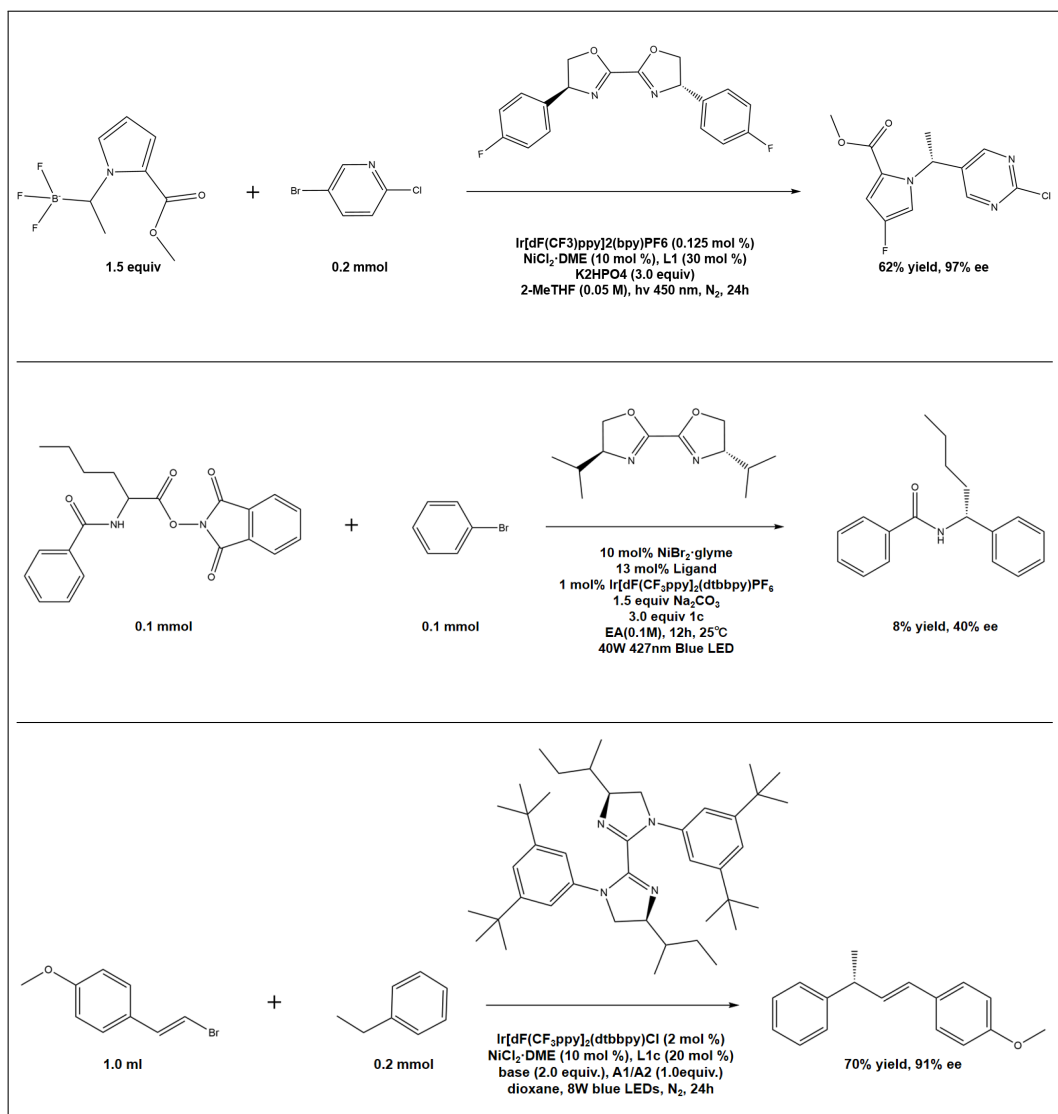


Figure 4: Three reactions in external dataset.

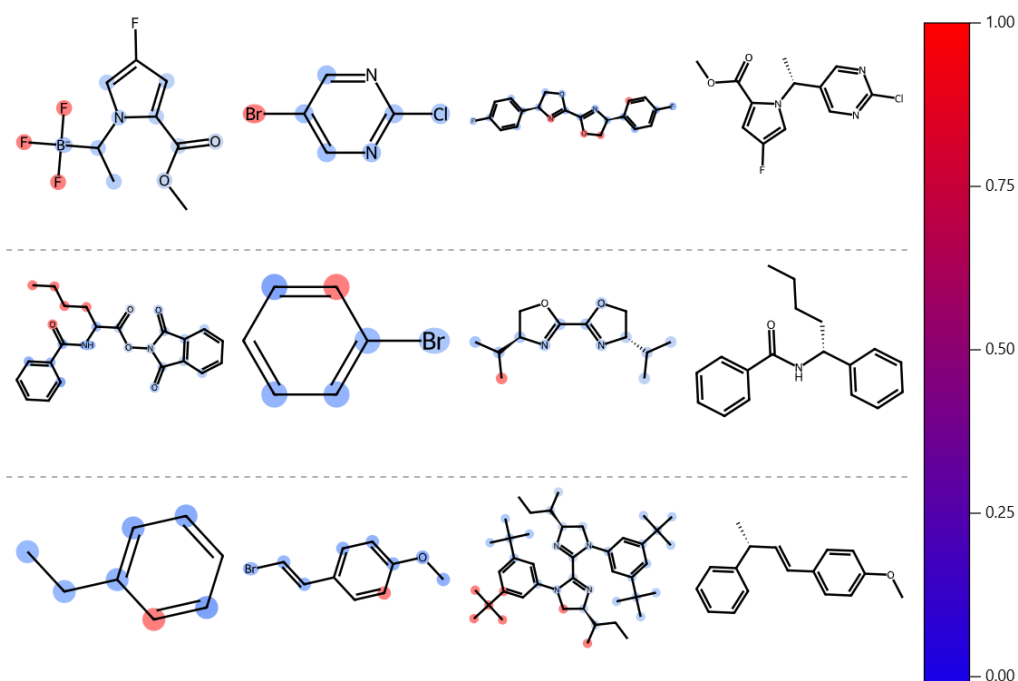


Figure 5: Atom-level model-derived saliency visualization. Each row corresponds to one reaction and displays, from left to right, reactant 1, reactant 2, catalyst, and product. The color scale represents relative nonnegative SAGPool node saliency (red: higher; blue: lower)

within the evaluated asymmetric catalysis reaction spaces. This approach provides a computational framework for enantioselectivity prediction within the evaluated reaction spaces. Additionally, we conducted comprehensive evaluations and ablation studies of CHIMERA on two curated chiral catalysis datasets. Experimental results highlight CHIMERA’s potential for enantioselectivity prediction in asymmetric catalysis.

Beyond predictive performance, CHIMERA provides model-derived atom-level saliency visualizations through the SAGPool attention mechanism, enabling chemists to inspect molecular regions receiving different levels of model attention. These saliency patterns may help identify molecular regions relevant to the model prediction and support chemically informed hypothesis generation.

Despite its strengths, CHIMERA still has several areas for improvement. Although the model effectively integrates molecular fingerprints, graph representations, and motif-based structures, incorporating more fine-grained physicochemical information remains an important challenge for improving the identification of key reactive sites and the interpretation of underlying reaction mechanisms. In addition, the current SAGPool-based saliency analysis highlights atom-level regions associated with the predicted magnitude of enantioselectivity, but it does not explicitly indicate which molecular regions favor the formation of a specific enantiomeric product, such as the *R* or *S* configuration. This limitation arises from the definition of the prediction target in the present datasets, where ee is calculated from the absolute difference between the two enantiomers and therefore does not retain directional information about the major product configuration. Future work will require signed ee values or consistently annotated absolute configurations of the major products, together with more explicit three-dimensional stereochemical representations, to enable prediction and interpretation of the sense of enantioselectivity. Furthermore, with the continuous accumulation of high-quality reaction data, future studies can incorporate a broader range of catalysts and reaction conditions into model training, thereby expanding the algorithm’s applicability. Supported by larger-scale and more heterogeneous datasets, the adaptability and robustness of CHIMERA in handling complex reaction systems are expected to be further enhanced.

5. Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

6. CRediT authorship contribution statement

Zongyong Hu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Jiebin Su: Methodology, Software, Formal analysis, Investigation, Validation, Data curation, Visualization, Writing – original draft. Jiaying Liu: Investigation, Validation, Resources, Data curation. Chunjin Wei: Validation, Formal analysis, Visualization, Writing – review & editing. Cheng Wang: Conceptualization, Supervision, Resources, Writing – review & editing, Project administration, Funding acquisition. Da Zhou: Conceptualization, Supervision, Writing – review & editing, Project administration, Funding acquisition.

7. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve the language and readability of the manuscript. After using this tool, the authors carefully reviewed and edited the content as needed and take full responsibility for the content of the published article.

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9. Data and Software Availability

The datasets and source code for the CHIMERA framework are openly available at <https://github.com/Y-bit8/CHIMERA>.

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